

CYBERSECURITY FUNDAMENTALS & ETHICAL HACKING BASICS

Internship Task 1

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Ethical Hacking & Cybersecurity

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1. INTRODUCTION

In today's digital world, computers, smartphones, cloud services, and online platforms store a huge amount of sensitive information. As technology grows, cyber threats are also increasing. Cybersecurity is the practice of protecting systems, networks, and data from unauthorized access, attacks, and damage.

Cybersecurity helps individuals and organizations maintain privacy, protect valuable information, and ensure smooth business operations. Without proper security measures, cybercriminals can steal data, disrupt services, and cause financial losses.

The primary goal of cybersecurity is to protect information systems from threats while ensuring data remains secure and accessible to authorized users.

Objectives of Cybersecurity

- Protect sensitive information
- Prevent cyber attacks
- Ensure data privacy
- Maintain business continuity
- Reduce security risks
- Protect digital assets

2. CYBERSECURITY FUNDAMENTALS

Cybersecurity consists of technologies, processes, and practices designed to protect systems and networks from attacks.

Cybersecurity focuses on:

Network Security

Protecting computer networks from unauthorized access and attacks.

Information Security

Protecting sensitive data from disclosure and modification.

Application Security

Securing software applications against vulnerabilities.

Cloud Security

Protecting cloud-based systems and services.

Endpoint Security

Securing devices such as computers, laptops, and smartphones.

Benefits of Cybersecurity

- Prevents data breaches
- Protects customer information
- Ensures system reliability
- Builds trust among users
- Reduces financial losses

3. ETHICAL HACKING

Ethical hacking is the authorized process of identifying vulnerabilities in computer systems and networks. Ethical hackers use the same techniques as malicious hackers but with permission and legal authorization.

The main purpose of ethical hacking is to discover weaknesses before cybercriminals can exploit them.

Importance of Ethical Hacking

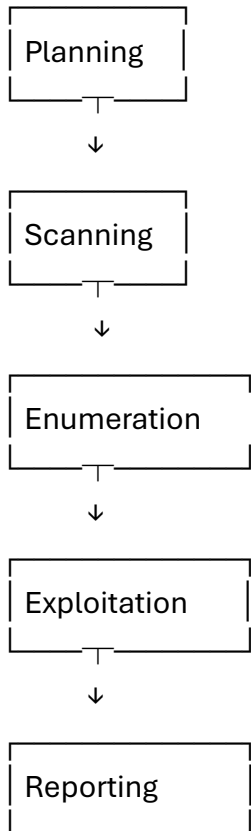
- Detects vulnerabilities
- Prevents cyber attacks
- Improves system security
- Protects sensitive information
- Helps organizations comply with security standards

Responsibilities of Ethical Hackers

- Vulnerability Assessment
- Penetration Testing
- Security Auditing
- Risk Analysis

- Security Reporting

Ethical Hacking Process Diagram



4. TYPES OF HACKERS

Hackers can be categorized based on their intentions and authorization.

White Hat Hackers

White Hat Hackers are ethical hackers who work with permission to improve security.

Characteristics:

- Legal and authorized
- Improve security
- Protect organizations

Examples:

- Penetration Testers
- Security Analysts

Black Hat Hackers

Black Hat Hackers perform illegal activities to gain unauthorized access to systems.

Characteristics:

- Unauthorized access
- Malicious intent
- Data theft

Examples:

- Cybercriminals
- Ransomware operators

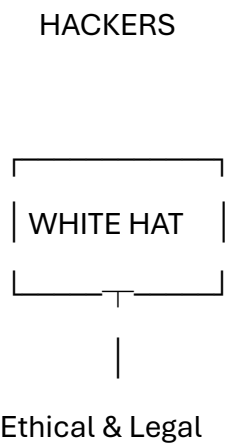
Grey Hat Hackers

Grey Hat Hackers operate without permission but usually do not have malicious intentions.

Characteristics:

- No authorization
- Ethical concerns
- Report vulnerabilities

Types of Hackers Diagram





No Permission



Illegal Activities

5. CIA TRIAD

The CIA Triad is the foundation of cybersecurity and consists of three essential principles:

1. Confidentiality
2. Integrity
3. Availability

Confidentiality

Confidentiality ensures information is accessible only to authorized individuals.

Examples:

- Passwords
- Encryption
- Access Control

Integrity

Integrity ensures information remains accurate and unchanged.

Examples:

- Hashing
- Digital Signatures
- Data Validation

Availability

Availability ensures systems and information remain accessible when needed.

Examples:

- Data Backups
- Disaster Recovery
- Load Balancing

6. COMMON CYBER THREATS

Cyber threats are activities that can damage, steal, or compromise digital systems.

Malware

Malware is malicious software designed to harm systems.

Types:

- Virus
- Worm
- Trojan Horse
- Spyware

Impact:

- Data loss
- System corruption
- Unauthorized access

Phishing

Phishing is a social engineering attack that tricks users into revealing sensitive information.

Examples:

- Fake Emails
- Fake Websites
- SMS Fraud

Impact:

- Credential theft
- Identity theft
- Financial loss

Ransomware

Ransomware is malware that encrypts data and demands payment for recovery.

Examples:

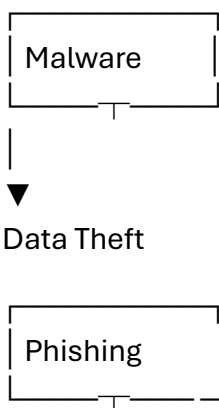
- WannaCry
- Ryuk
- Conti

Impact:

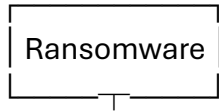
- Data encryption
- Business disruption
- Financial losses

Common Cyber Threats Diagram

COMMON CYBER THREATS



▼
Credential Theft



Data Locking

7. IMPACT OF CYBER THREATS

Cyber attacks can have severe consequences for organizations and individuals.

Threat	Impact
Malware	System damage and data theft
Phishing	Credential theft and fraud
Ransomware	Data encryption and financial loss
Spyware	Privacy violations
Worms	Network disruption

Organizational Impact

- Financial loss
- Reputation damage
- Service interruption
- Legal consequences

Individual Impact

- Identity theft
- Privacy breach
- Financial fraud

- Account compromise

8. PREVENTION TECHNIQUES

Organizations should implement multiple security measures to reduce cyber risks.

Malware Prevention

- Install antivirus software
- Update operating systems
- Avoid suspicious downloads

Phishing Prevention

- Verify email senders
- Avoid unknown links
- Enable Multi-Factor Authentication

Ransomware Prevention

- Maintain regular backups
- Update software frequently
- Train employees on cybersecurity awareness

Security Layers Diagram

SECURITY LAYERS

User Awareness

|



Strong Passwords

|



Multi-Factor Authentication



Antivirus



Firewall



Backups